Retrieval-Augmented LLMs for ETD Subject Classification

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***Abstract*—Electronic Theses and Dissertations (ETDs) constitute a vital part of the global scholarly record, but require accurate subject classification for effective archival management and retrieval. The computational challenges of processing thousands of ETDs annually, and implementing an effective subject classification within it exemplify one of the core issues in modern digital repositories. Current practices rely on author-assigned labels that need to be manually verified, while automated approaches using LLMs struggle with document length constraints and interdisciplinary research categorization. This paper presents a two-stage pipeline that addresses these challenges. First, we develop a novel summarization framework that extracts keywords from abstracts, generates targeted questions, and synthesizes answers from full document text through BM25 retrieval and retrieval-augmented generation (RAG). This approach helps create enriched representations that capture methodological nuances beyond what abstracts alone provide. Second, we implement reasoning-guided prompt engineering that embeds explicit disambiguation logic for frequently misclassified categories, particularly at disciplinary boundaries. Our approach demonstrates consistent improvements over baseline methods, with particular gains for interdisciplinary research and commonly confused category pairs. Analysis reveals that systematic misclassification patterns where application domains overshadow methodological contributions can be effectively addressed through structured prompting that prioritizes disciplinary markers and methodological approaches. This work provides both a practical solution for ETD classification and valuable insights to help improve LLMs' ability to distinguish between similar academic fields. Beyond classification, these generated summaries could prove valuable for enhancing search and indexing in digital repositories, supporting literature reviews, and enabling quick understanding of lengthy dissertations.**

***Index Terms*—archival records, subject classification, natural language processing, large language models, summarization, prompt engineering, natural language generation, digital libraries, computational archival science**

I. Introduction

Electronic Theses and Dissertations (ETDs) form one of the largest and most diverse collections of scholarly knowledge. They capture emerging research across disciplines, yet their discoverability critically depends on accurate subject classification within institutional repositories. In practice, this process remains inconsistent and fragile. Author-assigned subject labels—provided at deposit time—often conflate departmental affiliation with disciplinary focus, omit methodological context, or misrepresent interdisciplinary work. Manual curation is infeasible at scale: large universities process thousands of ETDs each year, and librarians face growing demands for metadata quality assurance under constrained resources. The result is a persistent misalignment between the conceptual structure of research and its archival representation—hindering discovery, citation, and cross-domain synthesis. This challenge reflects a fundamental issue in computational archival science. ETDs are quintessential archival records—permanent scholarly documents embodying core principles of provenance (institutional origin), original order (chapter structure), and evidential value (documenting research processes and outcomes). Their computational management exemplifies key CAS challenges: automating accurate description, ensuring metadata quality across thousands of records, and enabling discovery within complex classification systems. Addressing these issues is integral to developing automated methods that improve metadata generation, classification, and retrieval. As repositories increasingly adopt computational workflows for curation, automated classification, especially early in the ingestion pipeline, helps ensure that ETD collections remain accurate, discoverable and usable over time.

Large Language Models (LLMs) have recently been explored as a scalable approach to subject classification, offering the capacity to process large document collections and apply consistent criteria. However, empirical evaluations reveal recurring limitations. LLMs exhibit systematic biases in their classification decisions, relying heavily on lexical similarity while underrepresenting methodological and theoretical context [1]. They struggle with interdisciplinary research where multiple plausible categorizations exist and often

conflate closely related fields that share vocabulary but differ in epistemic orientation. The opacity of these models' decision processes further complicates the identification and correction of systematic errors, particularly those involving disciplinary boundary confusion.

A key difficulty arises from the level of information available for classification. Full ETD texts contain rich methodological detail necessary for accurate categorization but are typically hundreds of pages long—beyond the practical processing limits of most models. In contrast, titles and abstracts are computationally manageable but often omit critical information regarding methodology, theoretical grounding, and research context. This creates an inherent trade-off between computational feasibility and classification accuracy.

To address these challenges, we design a two-stage framework that separates information extraction from classification. The first stage performs retrieval-augmented summarization, selecting relevant passages from full-text ETDs and synthesizing them into concise representations that preserve methodological and conceptual content. The second stage performs reasoning-guided classification through LLMs, using prompts that incorporate disambiguation rules derived from systematic analyses of frequently confused categories.

In short, this study makes the following contributions:

- **A novel retrieval-augmented summarization pipeline:** We develop a summarization pipeline that extracts and synthesizes methodological details from ETDs to create concise representations for classification.
- **Reasoning-guided classification:** We design a prompt-based classification framework that applies explicit disambiguation rules to reduce confusion across closely related disciplines.
- **Comparative evaluation:** We provide a comparative analysis of the framework's performance against Machine Learning and fine-tuned transformer-based baselines, as well as results with different prompting strategies.

II. Related Work

The **classification** of scholarly long-form documents, such as Electronic Theses and Dissertations (ETDs), remains a core challenge in digital library and information retrieval research. Effective classification depends fundamentally on the accuracy and richness of their descriptive metadata, which underpins discoverability within archival systems. Over time, archival practice itself has expanded beyond traditional curation and preservation toward computationally supported methods for description, access, and analysis. This shift has brought archival science into closer alignment with information and computing research, emphasizing scalable, data-driven approaches to organizing complex scholarly materials. Within this broader context, advances in ETD classification align with the broader shift toward computational methods for archival that enhance metadata quality and facilitate discovery in digital repositories.

Researchers have long explored automated approaches to improve the discoverability of ETDs through metadata extraction. One early effort focused on identifying bibliographic elements such as titles and authors from scanned documents, demonstrating that textual and visual cues could together provide reliable descriptive metadata as shown by Choudhry et al. [2]. A subsequent challenge has been the accurate subject categorization of ETDs. Recognizing that conventional NLP models struggle with book-length texts, Jude [3] proposed chapter-level classification using custom embeddings and extractive summaries, significantly improving machine learning baselines such as Support Vector Machines and Random Forests. More recent work by Banerjee et al. [4] fine-tuned transformer architectures, including BERT and SciBERT, on ETD-specific corpora, showing that models adapted to academic discourse yield substantial gains for subject classification and summarization. Recent metadata research continues to advance these computational approaches. D'Souza et al. [5] organized the LLMs4Subjects shared task, demonstrating effective LLM-based automated subject tagging for technical library records using the GND taxonomy. Top-performing systems employed ensemble approaches, synthetic data generation, and multilingual models, achieving over 50% recall in subject prediction. These advances collectively reflect a convergence between archival description and computational modeling—advancing both the precision of metadata generation and the broader goals of discoverability and interoperability within digital repositories.

Recent advances in **language modeling** have substantially expanded the capabilities of natural language understanding and generation systems. Modern large language models (LLMs) are grounded in the transformer architecture [6], whose self-attention mechanism enables efficient contextual reasoning across long sequences. These models demonstrate strong generalization across tasks including classification, question answering, and summarization, enabling new applications in domains requiring the interpretation of long-form academic text. Within ETD classification research, this evolution represents a clear methodological shift from feature-based machine learning models toward reasoning-oriented and generative frameworks. Traditional classifiers such as Support Vector Machines (SVMs) and Random Forests were first used for chapter-level subject categorization of ETDs [3], while later studies extended these approaches using fine-tuned transformer models to better capture contextual semantics [4]. Building on this progression, recent work has shown that prompt-engineered and instruction-tuned models can rival or surpass supervised baselines, particularly in settings with limited labeled data [7]. Banerjee et al. [8], [9] demonstrated that LLM-based chapter-level classification enhances the discoverability and interdisciplinary linkage of ETDs by leveraging their ability to capture nuanced methodological and conceptual distinctions. Further studies [10], [11] indicate that both document length and model architecture significantly influence performance in long-form classification tasks. Collectively, these developments illustrate how the integration of prompt-guided reasoning and large-scale pretrained models advances automated metadata generation and improves access

to complex academic corpora.

Summarization offers a means to enrich the limited metadata traditionally available for ETDs. Titles and abstracts alone often fail to capture the methodological and disciplinary depth necessary for accurate classification, as they emphasize outcomes while omitting theoretical frameworks, research design, and analytical approaches. This lack of contextual detail can obscure interdisciplinary connections and lead to systematic misclassification [1]. Research on scholarly summarization has predominantly focused on enhancing information retrieval and content accessibility within academic repositories. Early efforts were largely extractive, ranking salient sentences based on surface-level features or graph centrality measures; while effective for brevity, such approaches often failed to capture methodological nuance or interdisciplinary perspective, as noted by Gana et al. [12]. Subsequent work transitioned toward neural abstractive models capable of synthesizing structured or extended abstracts encompassing motivation, methods, results, and implications of research. Transformer-based summarization systems such as BART, T5, and GPT variants have been employed for both single- and multi-document summarization, demonstrating substantial improvements in fluency and coherence [13]. Nonetheless, ETDs present unique challenges due to their extreme length and internal heterogeneity. Context window limitations, positional bias, and loss of mid-document coherence hinder the summarization of long-form academic works where methodological content is dispersed across chapters. Recent research has explored hybrid and hierarchical strategies to overcome these constraints. Alazhar et al. [14] integrated topic modeling with LLMs to analyze limitations sections across large scholarly corpora, while Ou et al. [15] proposed a hierarchical merging framework that dynamically combines local and global context during summary generation. Other approaches leverage entity extraction, scientific claims, or citation graphs to produce concept-centered summaries that improve retrieval and semantic indexing.

This work builds on these developments by situating summarization within a broader classification pipeline for ETDs. Our framework adopts a bottom-up question–answer-based strategy to construct summaries that emphasize disciplinary signals. The process begins with the extraction of salient keywords from each abstract, followed by the generation of targeted questions focused on the underlying concepts. Answers to these questions are then produced through the retrieval of relevant document segments and the synthesis of their content into coherent paragraphs. This design is inspired by question–answer-based summarization frameworks such as that of Hsu et al. [16], who demonstrated that structured Q&A generation can effectively integrate extractive and abstractive reasoning to yield informative and fluent summaries. The resulting summaries in our system serve as enriched, context-sensitive representations of each ETD, demonstrating scalable methods for generating enhanced metadata—supporting future efforts in indexing, discoverability, and information access within digital repositories.

III. Dataset and Taxonomy

Our dataset is a balanced subset selection of 2,100 ETDs from initial 9,218 ETDs obtained from the Virginia Tech institutional repository, collected in 2020. Each ETD record includes title, abstract, author, degree, and subject labels, and is cross-linked to its corresponding entry in the ProQuest Dissertations & Theses (PQDT) Global database. The linkage was established through a combination of automated and manual verification based on author names, titles, and publication dates.

PQDT organizes its collection using a hierarchical subject taxonomy with four levels and over 400 leaf-level academic disciplines. For this study, we adopted the most granular level of classification to represent each dissertation's disciplinary focus. These author-assigned categories serve as reference labels for all classification experiments.

To address class imbalance and ensure transparency, we ranked all 432 PQDT leaf-level categories by frequency and selected the top 28 most represented. To prevent an overwhelming majority of education-related disciplines and since education was not our primary focus, eight categories containing "education" (e.g., "Higher Education," "Adult Education") were merged into a single "Education" label and randomly subsampled to match other category sizes. This resulted in the final 21 categories in our corpus, spanning traditional engineering disciplines (mechanical, electrical, aerospace, chemical, civil, industrial, computer, and biomedical engineering), natural sciences (materials science, environmental science, molecular biology, organic chemistry, statistics, ecology, and forestry), computational fields (computer science and mathematics), and social sciences (marketing, education, public administration, and occupational psychology). The resulting dataset provides a controlled yet diverse benchmark for subject classification, capturing disciplinary variation across both STEM and non-STEM domains.

IV. Methodology

We formulate ETD classification as a multi-class prediction task in which each document is assigned to one of 21 subject categories. The disciplinary landscape is inherently complex, as many fields share analytical methods and theoretical foundations despite distinct intellectual boundaries. To address these overlaps, our methodology incorporates systematic extraction of disciplinary indicators and explicit disambiguation rules for frequently confused category pairs. As shown in prior work [1], relying solely on titles and abstracts provides insufficient information for distinguishing closely related disciplines, particularly when methodological approaches intersect across domains.

To address these limitations, we develop a two-stage pipeline that first generates document summarization through a RAG (retrieval augmented generation) pipeline employing abstract and full text, then performs reasoning-guided classification through LLM. This methodology stems from our observation that accurate disciplinary classification not only requires an understanding of what the research accomplishes,

but also how it accomplishes it. These specific methodological approaches, theoretical frameworks, and technical implementations serve as disciplinary signatures. These signatures drive the sophisticated prompt engineering in the second stage to reason about disciplinary boundaries using the enriched summary representations created in the first stage.

A. Summary Generation

Electronic Theses and Dissertations (ETDs) are long-form academic documents, typically spanning 50 to 300 pages, and thus present substantial computational challenges for language model–based summarization. Their length generally exceeds the context window capacity of current large language models (LLMs), making full-document processing infeasible and prone to information dilution. To address this limitation, we employ a retrieval-augmented generation (RAG) framework that combines targeted extraction with controlled abstraction. The retrieval stage applies the BM25 algorithm to identify document segments most relevant to disciplinary keywords derived from the abstract, ensuring that selected content captures core methodological and conceptual elements while reducing the document to approximately 5–10% of its original length. The subsequent abstractive stage synthesizes these retrieved segments into coherent summaries that retain both explicit research findings and implicit methodological characteristics critical for accurate subject classification.

1) Stage 1: Keyword-Guided Question Synthesis:

a) Disciplinary Keyword Identification: The initial phase employs LLM to extract salient keywords and concepts from ETD abstracts through structured prompting. This approach departs significantly from traditional methods like TF-IDF or RAKE, which rely on statistical properties. While these methods identify frequently occurring terms, they miss conceptual relationships, implicit domain knowledge, and semantically important terms that appear infrequently but carry substantial disciplinary significance.

The extraction process instructs LLM to identify key concepts, methodologies, technologies, locations, organizations, and technical terms from abstracts through a straightforward prompt targeting named entities and important keywords. The model adapts to different fields by leveraging its training on academic discourse rather than using predefined extraction rules.

Extraction typically yields 10 to 30 keywords per abstract, with interdisciplinary research producing larger sets from multiple fields. Post-processing filters noise while preserving academic terminology, including standard abbreviations and domain-specific terms indicating disciplinary alignment.

b) Keyword-Driven Question Generation: The keywords identified in the previous step are used to generate targeted questions that guide focused exploration of each ETD. By asking questions centered on these concepts, the model can later retrieve concise, relevant answers that capture the methodological and conceptual core of the work. This approach ensures that the resulting questions remain specific to the disciplinary focus of the ETD and support accurate, context-aware summarization.

```
prompt = """Perform named entity recognition
    and keyword extraction on the following
    abstract.
Extract domain-specific terms across these
    categories:

- Core concepts and theoretical frameworks
- Research methodologies and analytical
    techniques
- Technologies and computational tools
- Geographic locations and institutional
    affiliations
- Technical terminology and discipline-
    specific vocabulary

Abstract: {abstract}

Return only the JSON array without additional
    text.
Format: ["term1", "term2", "term3", ...]"""
```

Fig. 1. Prompt for keyword extraction from abstracts

To maintain consistency and efficiency, the LLM is instructed to generate exactly ten questions per abstract, each grounded in the extracted keywords. These questions probe different aspects of the research—its objectives, methodology, findings, and application domain—ensuring structured coverage of the work. The process draws inspiration from educational assessment, where well-designed questions elicit meaningful understanding across multiple dimensions. By converting each abstract into a concise set of guided inquiries, the system establishes a foundation for focused retrieval and summarization from the full document text.

The generation prompt explicitly connects the abstract content with the extracted keywords, creating a bridge between identified concepts and their exploration:

The prompt explicitly connects abstracts with extracted keywords, ensuring questions probe specific methodologies and concepts. For instance, when *neural networks* and *medical imaging* appear as keywords, generated questions explore *how neural networks are applied to medical imaging*, *what advantages they provide*, and *what specific imaging problems they address*.

The generic prompt framework targets universal elements such as domain identification, methodological approaches, and practical applications—while using keywords to generate field-specific terminology rather than generic queries. The non-expert answerability constraint enforces clarity and prevents complex formulations that might obscure research characteristics.

The quality of generated questions provided an indirect indication of the relevance of extracted keywords. This feedback relationship helped identify occasional cases of keyword extraction failure in our work. To further assess reliability,

```
prompt = """Generate exactly 10 questions that
     someone trying to understand this
    research would ask.
Focus on: the domain of the research, its
    methodology (use keywords as the main
    guide),
findings, and the application areas and
    beneficiaries of the research.

Write simple questions that can be answered by
     a non-expert just by reading from the
    abstract.

Abstract: {abstract}

Key concepts and entities identified: {
    keywords_str}

Make sure to generate questions that explore
    these key concepts
and how they relate to the research."""
```

Fig. 2. Prompt for question generation from research abstracts

we manually reviewed 10% of the generated keywords and questions across ETDs in each subject category. Overall, the outputs were found to be relevant and consistent with expectations, though in some instances the questions were overly broad or abstract. Such cases were attributed to inherent variability in LLM generation, and given that subsequent answer generation was context-constrained, these deviations were not expected to impact overall summarization quality.

2) Stage 2: Answer Generation using RAG: The answer generation stage employs Retrieval-Augmented Generation (RAG) to efficiently summarize long ETD documents while preserving disciplinary context. Full ETDs often exceed 40,000 tokens, making direct processing computationally infeasible for current language models. RAG mitigates this by first retrieving only the most relevant document segments and then generating grounded, context-specific answers to the previously formulated questions. This approach reduces the effective input size by approximately 90%, ensuring that generated summaries are derived from authentic ETD content rather than model priors.

a) Document Selection for Context Creation: Keywords extracted in the previous stage serve as queries for retrieving semantically relevant sentences from each ETD. Retrieval is performed using the BM25 algorithm, which scores sentences based on term frequency and inverse document frequency:

$$\text{Score}(s,k) = \sum_{t \in k} \text{IDF}(t) \cdot \frac{f(t,s) \cdot (k_1 + 1)}{f(t,s) + k_1 \cdot (1 - b + b \cdot \frac{|s|}{\text{avgdl}})}$$

where the inverse document frequency (IDF) component is:

$$\text{IDF}(t) = \log\left(\frac{N - df(t) + 0.5}{df(t) + 0.5}\right)$$

```
prompt = """Answer these questions using ONLY
    the provided context.

IMPORTANT INSTRUCTIONS:
- Write complete, full sentences in your
    answers, not just phrases
- Make each answer coherent, readable, and
    reasonable
- Write at least a paragraph (4-5 sentences)
    for each answer
- Use ONLY information from the provided
    context
- If the answer is not in the context, write a
     paragraph explaining what information is
    missing
- Ensure answers flow naturally and provide
    comprehensive explanations
- DO NOT write answers that include the
    related literature or citations (
    publications, authors, years, etc.)

Context: {context}

Questions to answer:
{json.dumps(questions_to_answer, indent=2)}

IMPORTANT: Return ONLY a valid JSON object
    where keys are the exact questions and
    values are the paragraph-length answers.
"""
```

Fig. 3. Prompt for answer generation using retrieved context

Here, s represents a sentence from the document, k a keyword query, $f(t,s)$ the frequency of term t in sentence s, N the total number of sentences, $df(t)$ the number of sentences containing term t, and avgdl the average sentence length. The parameters $k_1 = 1.2$ and $b = 0.75$ control term frequency saturation and length normalization respectively.

For each keyword, the top 10 sentences are retrieved and deduplicated, yielding approximately 50–200 context sentences per document. The BM25 retrieval process queries the document using discipline-specific keywords, naturally drawing sentences from multiple sections. This preserves contextual continuity and captures both methodological and conceptual information required for classification.

b) Answer Generation from Retrieved Context: Each predefined question is then answered by the LLM using only the retrieved context. The prompt explicitly restricts generation to grounded content, preserving authentic disciplinary terminology and phrasing. Responses are required to form complete paragraphs of 4–5 sentences, ensuring grammatical and conceptual coherence. Citations and external references are excluded to focus on the research itself.

A near-deterministic `temperature` of 0.1 maintains consistency across runs, and the 10,000-token limit accommodates detailed responses for all questions. The resulting summaries provide concise yet comprehensive representations of each ETD, capturing methodological, theoretical, and conceptual

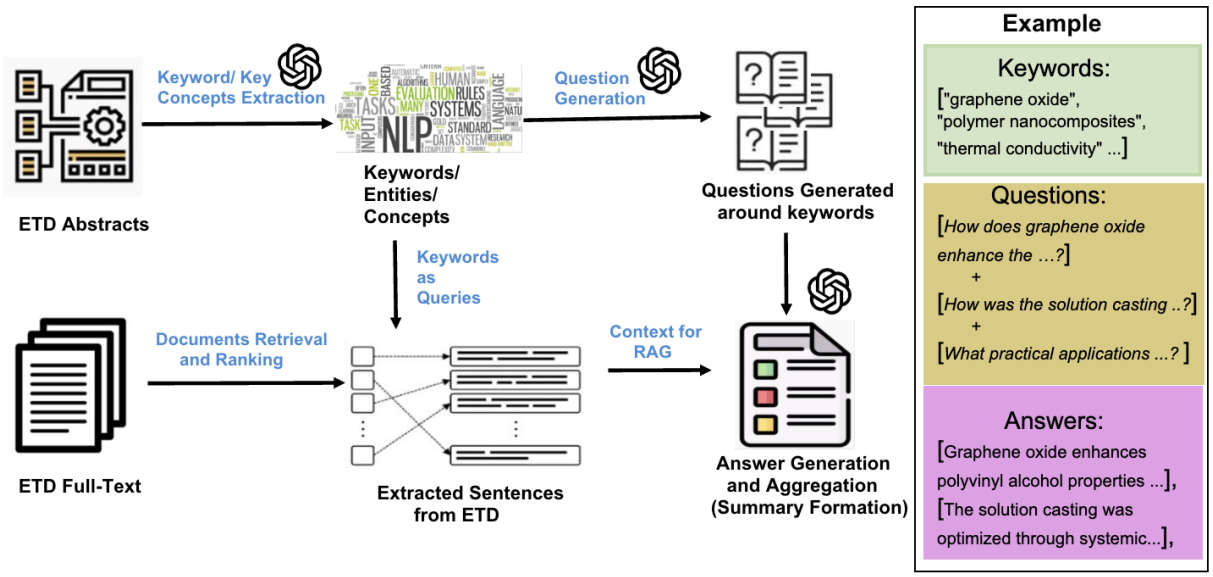


Fig. 4. Overview of the Summarization Pipeline

Require: Questions $Q = \{q_1, \ldots, q_N\}$, Keywords $K = \{k_1, \ldots, k_n\}$, Document D
Ensure: Answers $A = \{a_1, \ldots, a_N\}$
$A \leftarrow \emptyset$
$S \leftarrow$ SentenceTokenize(D)
$S_{\text{filtered}} \leftarrow$ FilterByLength$(S$, min=5, max=500$)$
Initialize BM25Index$(S_{\text{filtered}}, k_1$=1.2, b=0.75$)$
Context $\leftarrow \emptyset$
for all keyword k_i in K **do**
  scores $\leftarrow$ BM25Index.Score$(k_i, S_{\text{filtered}})$
  top_sentences $\leftarrow$ SelectTop10(scores, S_{filtered})
  Context $\leftarrow$ Context $\cup$ top_sentences
end for
Context $\leftarrow$ RemoveDuplicates(Context)
Prompt $\leftarrow$ "Answer using **ONLY** provided context..."
Response $\leftarrow$ LLM(Prompt, Q, Context, temp=0.1)
ResponseDict $\leftarrow$ ParseJSON(Response)
for all question q_i in Q **do**
  $a_i \leftarrow$ ExtractAnswer(ResponseDict, q_i, case_insensitive=True)
  if ValidateAnswer$(a_i$, min_sentences=4$)$ **then**
    Add a_i to A
  else
    Add DefaultAnswer(q_i) to A
  end if
end for
return A

Fig. 5. RAG-based Answer Generation

details critical for subsequent classification.

c) Final Summary Creation and Quality Assurance: The final summary, obtained by concatenating all generated answers, provides a richer and more informative representation than the abstract alone. It captures methodological details, domain-specific terminology, and interdisciplinary relationships while maintaining factual grounding through the retrieval-augmented generation (RAG) process.

To ensure reliability, both manual and automated quality control procedures were applied. A random subset of 15 ETDs from each subject category was manually reviewed to assess the coherence and topical relevance of the generated keywords, questions, and summaries. In addition, automated validation was performed on the final summaries, where outputs containing fewer than 20 sentences were automatically flagged and excluded from downstream processing. Approximately 7% of summaries were removed under this criterion. On average, each ETD required 90–100 seconds for complete processing, yielding concise yet comprehensive summaries that substantially improve classification accuracy—particularly for interdisciplinary and boundary-spanning research. These generated summaries also function as archival surrogates—condensed documentary representations that preserve essential intellectual content while facilitating discovery and assessment. This concept parallels "paradata" from archaeological documentation, where process-oriented metadata enriches understanding of primary materials without requiring full examination.

B. Classification

Building on prior analyses of LLM-based ETD classification [1], we observed that standard prompting approaches often resulted in systematic patterns of misclassification, particularly at disciplinary boundaries. Closely related fields—such as those within engineering, computational sciences, and social disciplines—were frequently conflated due to overlapping terminology and methodological similarities. To better understand this phenomenon and design better prompts for this issue, we conducted an error analysis of the ten lowest-performing categories, which collectively accounted for nearly 70% of overall misclassifications. Each classification output was accompanied by an LLM-generated rationale, allowing detailed inspection of the model's reasoning process. This qualitative examination revealed that surface-level lexical cues and application domains often overshadowed core method-

```
prompt = f"""Based ONLY on the following
    summary, classify this academic document
    into ONE of the provided categories.

Title: {title}
Summary: {summary}

Categories: {json.dumps(CATEGORIES)}

CATEGORY-SPECIFIC CLASSIFICATION RULES:

**aerospace engineering**:
  - Priority indicators:
      aircraft, spacecraft, satellites,
      rockets, UAVs, flight vehicles
  - Key combinations:
      combustion+hypersonic -> aerospace
  - Focus on:
      the SYSTEM (flight vehicle) not just
      the METHOD

**computer science**:
  - Priority indicators:
      new algorithms, computational models,
      software architectures
  - Key combinations:
      deep learning/ML+any application -> CS
  - Focus on:
      whether COMPUTATIONAL METHODS are
      central to the contribution

**statistics**:
  - Priority indicators:
      statistical modeling, estimation,
      inference, hypothesis testing
  - Focus on:
      DESIGN, ANALYSIS, and INTERPRETATION
      of DATA

[Additional categories truncated for space...]

STRATEGY:
1. Check for foundational discipline
    indicators
2. Look for compound keyword combinations
3. Identify focus on core methods vs.
    operational problems
4. Apply field-specific context rules"""}
```

Fig. 6. Classification prompt with disambiguation rules for disciplinary boundaries

ological contributions. For instance, research emphasizing machine learning techniques for biomedical or educational applications was frequently labeled according to the application field rather than the methodological focus. These findings motivated the development of a structured reasoning framework that explicitly guides the model through a discipline-aware decision process.

1) Reasoning-Guided Prompt Engineering: To address these systematic misclassifications, we developed an enhanced prompt that embeds disciplinary disambiguation logic directly into the classification process. The prompt incorporates category-specific rules including:

Priority Indicators: Discipline-specific terms that strongly indicate membership (e.g., "aircraft, spacecraft, UAVs" for aerospace engineering; "process design, reactor design" for chemical engineering).

Key Combinations: Co-occurrence patterns that distinguish disciplines (e.g., "combustion+hypersonic $\rightarrow$ aerospace" versus "combustion+process optimization $\rightarrow$ chemical engineering").

Focus Directives: Explicit rules about what to prioritize—computer science emphasizes computational/algorithmic contributions regardless of application domain, while biomedical engineering prioritizes biological/medical applications even when using standard engineering methods.

Disambiguation Rules: Specific criteria for frequently confused pairs—computer engineering requires hardware-software integration while computer science focuses on algorithms; electrical engineering is indicated by circuits, signals, and power systems even when software is present.

This structured prompt serves as a transparent reasoning scaffold rather than a hard-coded rule base, enabling flexible yet interpretable classification across diverse academic domains. Following prompt refinement, misclassification patterns at disciplinary boundaries were reduced, as shown by the improvements in evaluation metrics. The prompting process operates deterministically (temperature = 0.1, max_tokens = 20), ensuring consistency across model runs, and all predicted outputs are normalized through fuzzy string matching to align with the controlled 21-category taxonomy.

By integrating structured reasoning directly into the prompting process, this mechanism advances beyond prior heuristic classification approaches. It allows the LLM to articulate its decision path, align category selection with methodological intent, and improve the interpretability of automated subject labeling within large-scale academic repositories.

V. Experiments

A. Models and Configurations

The proposed reasoning-guided framework was evaluated across 21 academic subject categories using multiple input configurations and model variants. The primary objective was to examine how enriched document representations and structured reasoning instructions jointly affect classification performance when applied to long-form ETDs.

1) Primary Experimental Setup: All experiments employed `GPT-4.1-mini` (OpenAI API) as the inference model, selected for its strong performance-cost balance observed in prior work. It had demonstrated performance comparable to the larger `GPT-4.5-preview` model but at substantially lower computational cost. It also consistently outperformed open-source alternatives such as `LLaMA-3 (7B)` and `Mistral v0.3` on classification accuracy and consistency

TABLE I
COMPREHENSIVE CLASSIFICATION PERFORMANCE ACROSS ALL MODEL GROUPS AND CONFIGURATIONS

Group	Model / Configuration	Accuracy	Precision	Recall	F1
Our Method	Abstract + Summary + Reasoning	69.01%	**73.49%**	**69.00%**	**68.96%**
	Summary + Reasoning	68.05%	72.58%	68.05%	68.04%
	Abstract + Reasoning	67.86%	73.35%	67.85%	67.93%
	Abstract + Summary	66.48%	70.89%	66.49%	66.61%
	Summary Only	66.00%	70.17%	66.01%	66.13%
	Abstract Only	66.57%	71.35%	66.58%	66.65%
Prompt-based	GPT-4.1-mini (Few-shot)	**76.15%**	70.93%	66.83%	66.86%
	GPT-4.1-mini (Zero-shot)	75.94%	70.71%	65.18%	65.47%
	Mistral v0.3 (Zero-shot)	58.74%	67.92%	58.48%	56.77%
Fine-tuned	BERT	72.07%	58.95%	61.71%	58.75%
	LLaMA-3	68.15%	66.81%	67.95%	65.95%
	GPT-2	51.34%	49.94%	51.19%	47.08%
Traditional ML	SVM (TF-IDF)	72.10%	59.43%	61.75%	59.49%

across categories. Given this tradeoff between accuracy, stability, and cost-efficiency, `GPT-4.1-mini` was chosen as the primary inference model for all prompt-based configurations.The temperature was fixed at 0.1 to ensure deterministic and reproducible predictions while maintaining sufficient flexibility for borderline interdisciplinary cases.

Each prompt was composed of three core elements:

1) **Task Specification:** concise instruction directing the model to assign exactly one subject category.
2) **Category Space:** complete list of the 21 valid ProQuest subject categories.
3) **Reasoning Layer:** discipline-specific guidance and disambiguation rules derived from prior misclassification analysis.

2) Input Configurations: To systematically assess the contribution of different text representations, we designed three input settings:

- **TA (Title + Abstract):** baseline configuration using standard metadata available at deposit time.
- **TAS (Title + Abstract + Summary):** enriched input combining standard metadata with the retrieval-augmented summary produced in earlier pipeline stages.
- **S (Summary Only):** configuration using only the synthesized summary for classification.

For each configuration, two variants were evaluated:

- **Without reasoning:** model predicts the category based solely on textual cues and the category list.
- **With reasoning (ours):** model receives additional structured guidance mentioned earlier

This forms a controlled design setup (with/without reasoning), serving as an ablation study to isolate the effects of input elements and reasoning instructions.

3) Comparative Baselines: The proposed frameworks were evaluated against prior baselines and reference systems:

- **Prompt-based baselines:** zero-shot and few-shot prompting using GPT-4.1-mini, and Mistral v0.3. These prompts had no reasoning in the instruction, and only had title and abstract as input.
- **Fine-tuned models:** BERT-base-uncased [17], LLaMA-3 (7B) [18], and GPT-2 fine-tuned on the intitial 9,218 ETD titles and abstracts, for our 21 categories.
- **Traditional baseline:** Support Vector Machine (SVM) with TF–IDF features.

The open-source and traditional models serve as comparative baselines, providing reference points for quantifying the gains achieved through reasoning-guided prompting. The core contribution of this work, however, lies not in scaling model architectures or optimizing parameters, but in the systematic design and evaluation of structured reasoning prompts, supported by retrieval-augmented summaries that offer a more comprehensive and methodologically rich representation of each ETD than abstracts alone.

B. Implementation Details

Each ETD full text was divided into fixed-length chunks to manage input within the model's context window limitations. This segmentation was performed sequentially, processing a fixed number of tokens at a time without applying semantic boundary detection. From these chunks, the BM25 retrieval mechanism was applied to rank and retain only the most relevant portions (typically 5–10% of the total text) based on extracted disciplinary keywords. This ensured computational feasibility while maintaining representative content coverage.

The final input sequence—comprising selected chunks along with the title and abstract—was limited to 10,000 tokens, with the classification output restricted to 20 tokens to enforce concise, single-label predictions. All experiments were executed on two NVIDIA L40S GPUs (48 GB each) under consistent configurations to ensure stable performance and reproducibility across runs.

C. Comparative Performance

Table I summarizes the comparative performance across all configurations using macro-averaged accuracy, precision, recall, and F1-score.

VI. Results and Analysis

Our reasoning-guided framework achieved consistently strong performance across all tested configurations and models. Compared with traditional machine learning and fine-tuned transformer baselines, the reasoning-enabled LLM approach produced more stable and reliable results. The best configuration—combining abstracts and summaries with reasoning—achieved an overall accuracy of 69.01% and a macro-averaged F1-score of 68.96%, surpassing all non-reasoning variants by 2–3 percentage points on average.

A. Impact of Summary-Based Representations

Summaries provided a more informative and context-rich representation of ETDs than abstracts alone. When reasoning was applied, the benefit of summary-based input became more evident—accuracy increased to 68.05% for summary-only and 69.01% for the combined configuration. These results confirm that summaries capture methodological and contextual cues absent in abstracts, enabling more accurate differentiation between disciplines with similar research approaches.

B. Impact of Reasoning-Guided Prompting

Introducing explicit reasoning instructions improved performance in every configuration. Abstract-only accuracy rose from 66.57% to 67.86% (+1.29 pp), summary-only from 66.00% to 68.05% (+2.05 pp), and the combined inputs from 66.48% to 69.01% (+2.53 pp). Although GPT-4.1-mini (few-shot) achieved the highest raw accuracy (76.15%), its F1-score (66.86%) lagged behind our best reasoning-based configuration (68.96%). We would like to mention here that accuracy alone can be misleading in interdisciplinary classification, as it tends to favor dominant or unambiguous categories while obscuring performance on boundary-spanning classes. Even with a balanced dataset, it does not fully represent the model's capacity for consistent decision-making. Macro-averaged F1-score therefore offers a more rigorous measure of classification quality, emphasizing balanced precision and recall across all categories. Our approach maintained lower variance in per-category F1-scores (standard deviation 0.12 vs. 0.19), indicating greater balance and robustness.

C. Computational Efficiency

The end-to-end pipeline required approximately 90–100 seconds per ETD, including summarization and classification. Even so, this remains practical for institutional-scale processing of thousands of ETDs annually. Compared with manual metadata validation that may typically take 15–20 minutes per document, this pipeline represents a substantial efficiency gain while ensuring consistent, high-quality subject labeling across large academic repositories.

VII. Conclusion and Future Work

This study demonstrates that retrieval-augmented LLMs effectively address ETD subject classification challenges within computational archival science. Our two-stage pipeline achieves 69.01% accuracy by combining targeted summarization with reasoning-guided classification, surpassing baseline approaches while maintaining computational feasibility for institutional-scale processing. The framework successfully disambiguates interdisciplinary research and closely related fields, providing a practical solution for metadata quality assurance in digital repositories.

While the summaries developed in this study were introduced as intermediate representations for subject classification, they hold broader potential within digital repositories. Concise, methodologically informative surrogates for long-form ETDs can enhance search precision, improve browsing across disciplines, and support faster relevance assessment by researchers and librarians. Future work will focus on treating these summaries as independent metadata assets, evaluating their effect on retrieval quality and collection-level exploration in real repository systems.

Further refinement of the summarization process is needed to improve coverage and factual accuracy without increasing computational cost. Enhancements such as improved retrieval strategies, question generation, and omission detection could yield more reliable summaries. Equally important is systematic human evaluation—conducted with subject librarians and disciplinary experts—to assess summary quality, factual grounding, and usefulness for metadata enrichment. The framework may also be extended to other archival tasks such as keyword assignment and automated metadata validation. Collectively, these directions position retrieval-augmented summaries as a practical foundation for improving ETD classification and metadata generation, advancing the broader goals of accuracy, transparency, and efficiency in archival science.

Beyond these technical improvements, the methods developed here extend to broader computational archival science applications. The retrieval-augmented summarization pipeline could process diverse archival collections—government records, manuscript collections, corporate archives—where full-text analysis is computationally prohibitive. Similarly, our reasoning-guided classification framework offers a transferable approach for organizing large-scale archival corpora with overlapping categories. By validating these techniques on ETDs, this work provides scalable methods for automated description and arrangement applicable across archival domains, advancing the core CAS mission of enhancing access through computational innovation.

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